# Project Report

# Design and Simulation of a 72V to 12V DC-DC Buck Converter for Electric Vehicle Accessories

Power electronics – PUSL2089

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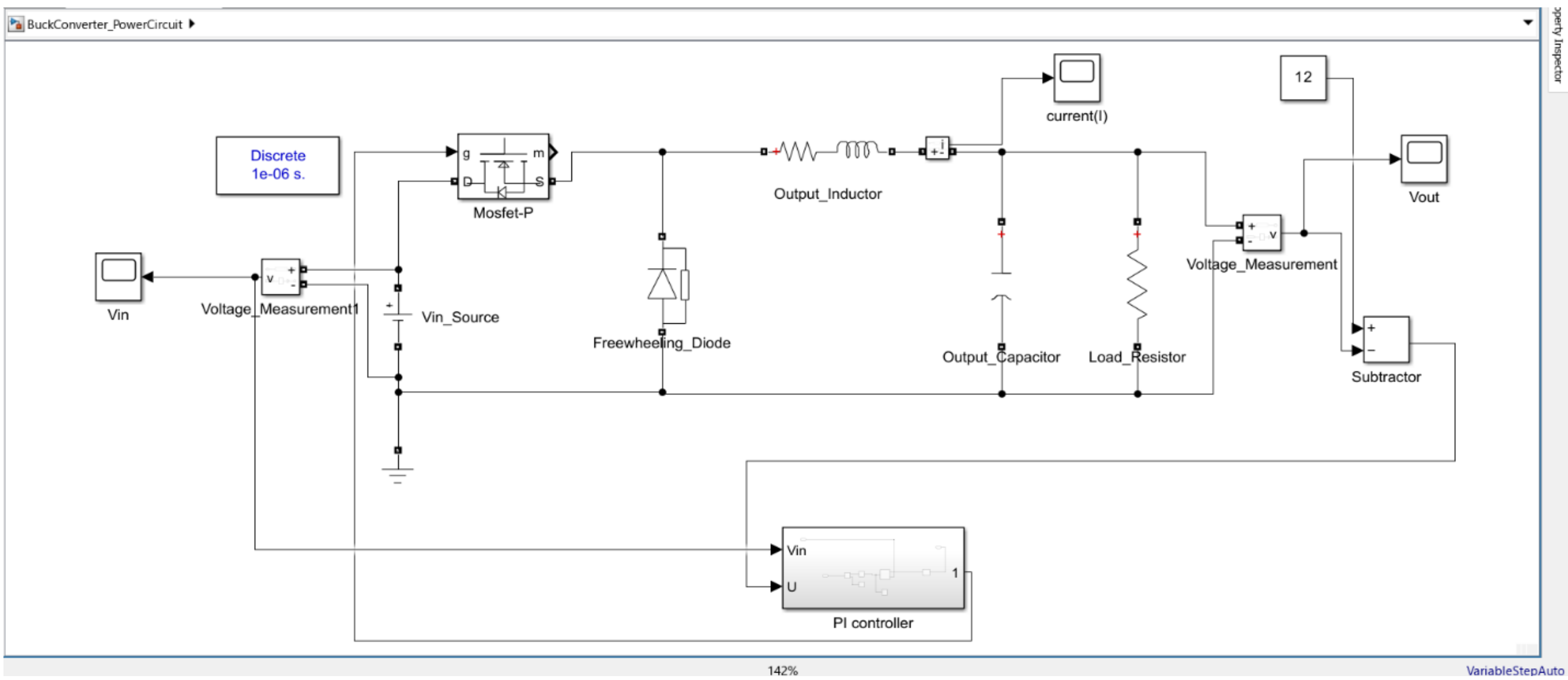
This project focuses on the design and simulation of a DC-DC buck converter that steps down a variable input voltage (60–90V) to a stable 12V output for electric vehicle  accessories. The converter was modeled in MATLAB/Simulink, incorporating a unique approach to duty cycle control and a tuned PI controller for voltage regulation. Key design equations, simulation scenarios, and performance analysis are discussed to justify component selection and control logic.

## Buck Converter Design Parameters

| **Component / Parameter** | **Value** | |
|---|---|---|
| Input Voltage | 60-90 | V |
| Output Voltage | 12 | V |
| Output Current-up to | 10 | A |
| Load Resistance | 1.2 | Ω |
| Switching Frequency | 100,000 | Hz |
| Timer Period | $10 \times 10^{-6}$ | s |
| Sample Time | $1 \times 10^{-6}$ | s |
| Inductor LL | 4 | mH |
| Inductor Resistance RLR_L | 1 | Ω |
| Capacitor CC | 220 | μF |
| Freewheeling Diode Forward Voltage | 0.9 | V |
| Diode Resistance | 0.01 | Ω |
| MOSFET On-Resistance | 0.01 | Ω |

| Component / Parameter | Value |
|---|---|
| MOSFET Internal FET Value | 0.1 |

## Simulink design overview




1. Vin_source

- Simulates the input battery voltage from an EV - 60V to 90V

2. MOSFET-P

- Controls when current flows through the inductor by act as a switch
- Turns ON and OFF at 100Khz signal on PWM from the PI controller block

3. Freewheeling_Diode

    - Provides a path for inductor current when the MOSFET is OFF To prevent sudden voltage spikes and allows continuous current flow

4. Output_Inductor

$$L = \frac{(V_{in} - V_{out}) \cdot D}{f_s \cdot \Delta I_L}$$

    - The inductor value is calculated to limit current ripple to 15% of the maximum output current – 4mH
    - It Stores energy and smoothens current flow to the load

5. Output_capacitor

$$C = \frac{I_{out} \cdot D}{f_s \cdot \Delta V_{out}}$$

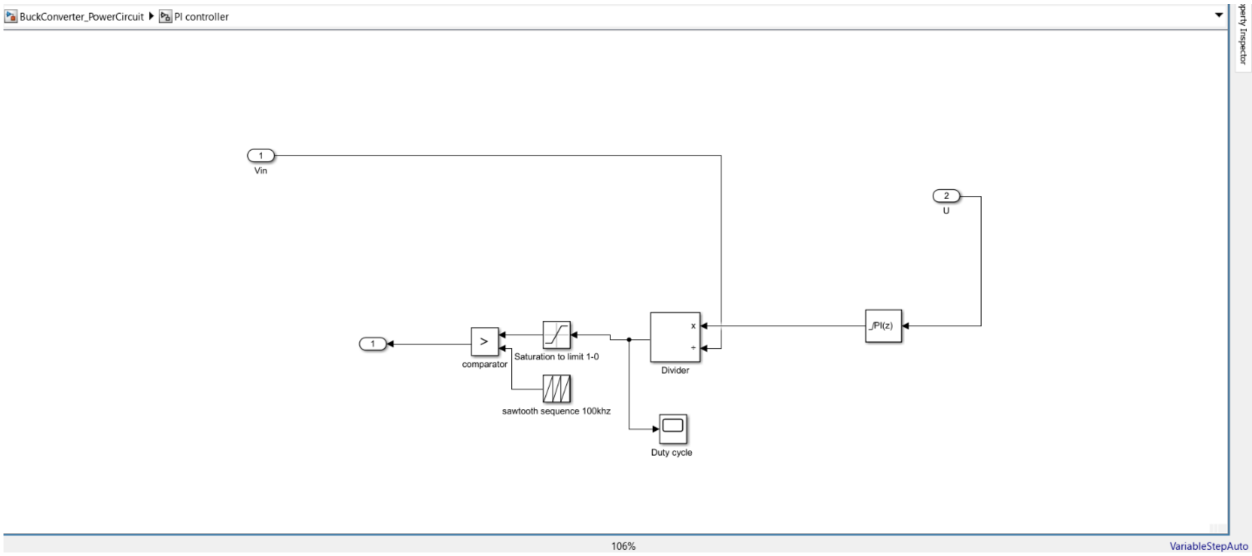
- Filters voltage ripple on the output side.
- Maintains steady 12V output despite switching noise

6. Subtractor

    - calculate the **error** between the desired output voltage and the actual output voltage. This error is what drives the PI controller

      e = Vref-Vout

## PI Controller Block




## 7. PI Controller

$$u(t) = K_p \cdot e(t) + K_i \int e(t)\, dt$$

- PI controller regulates the output voltage by dynamically adjusting the MOSFET’s duty cycle
- A sensor measures *Vout*. The measured *Vout* is subtracted from the reference-12V and The U control signal that tells how hard we need to correct the error

## 8. Divider

$$D = \frac{u(t)}{V_{\text{in}}}$$

- Since the PWM block expects a **duty cycle (ratio 0–1)**, we normalize the PI output by dividing it by the current input voltage

- This gives the **duty cycle (D)** for the PWM.

9. Sawtooth sequence 100Khz

- Generates a periodic rising waveform (0 to 1) used to compare with the duty cycle to create a PWM signal

10. Saturation

- Clamps the **duty cycle** to stay between 0 and 1

11. Comparator

- Compares the **duty cycle** from PI output (normalized between 0–1) with the sawtooth waveform to generate PWM

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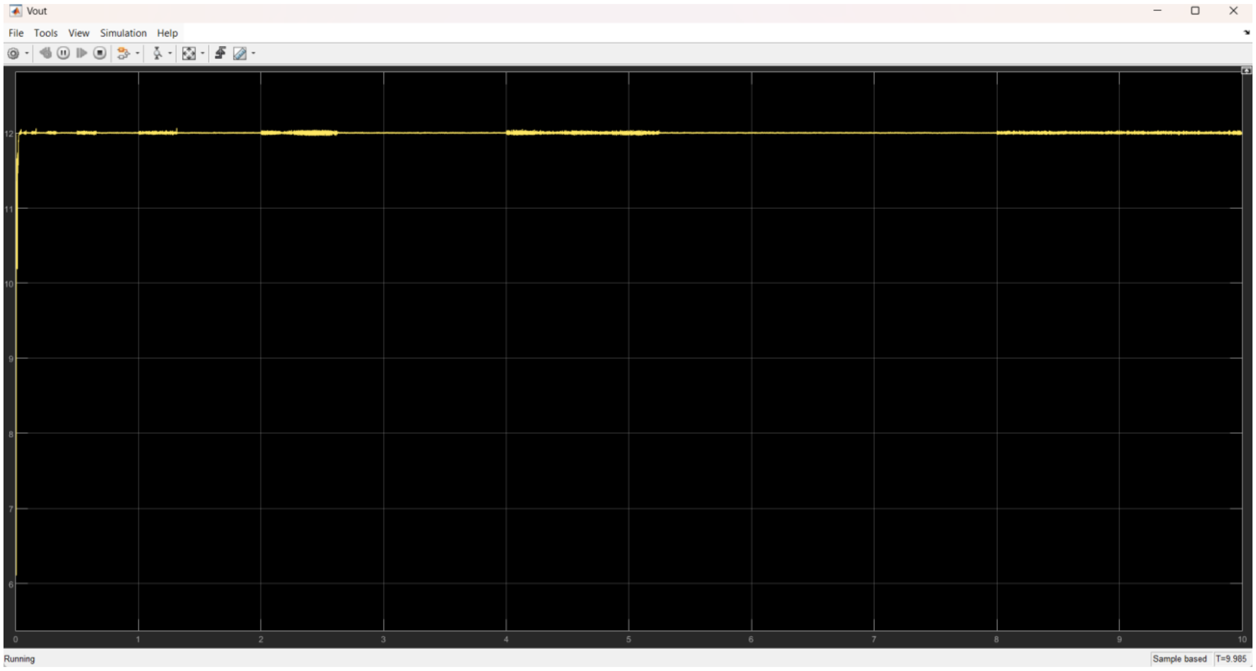
IF D > SAWTOOTH → OUTPUT = 1 (MOSFET ON)
ELSE            → OUTPUT = 0 (MOSFET OFF)
```

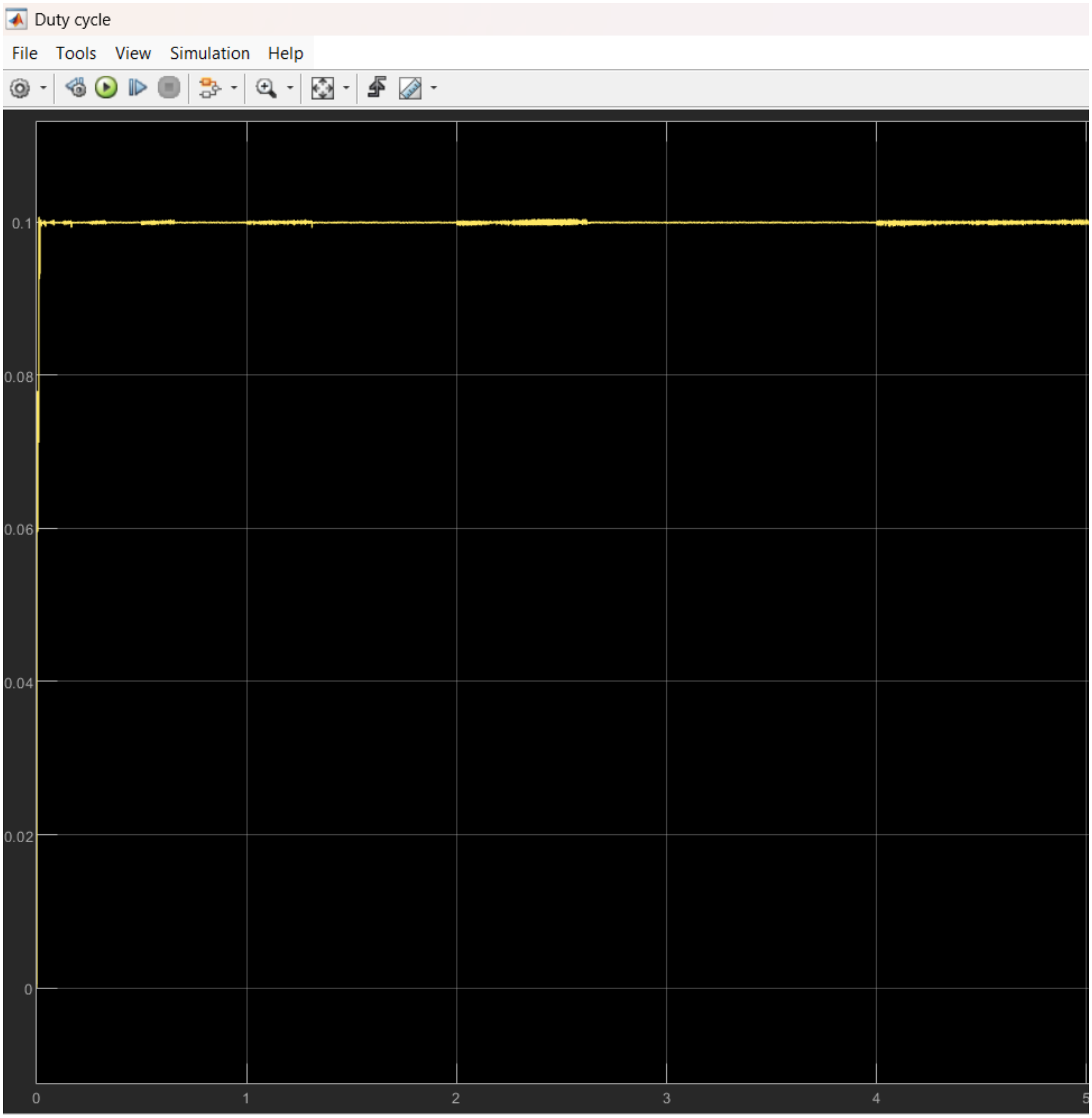
V out


Vout
File Tools View Simulation Help
12
11
10
9
8
7
6
0
1
2
3
4
5
6
7
8
9
10
Running
Sample based
T=9.985

I out


current(I)
File Tools View Simulation Help
Ready
Sample based
T=10.000

Duty Cycle – Vin(90V)



## Discussion

The converter successfully maintains a stable 12V output across all scenarios. The use of a dynamic duty cycle combined with a tuned PI controller ensured high performance.

The control system responded well to rapid input fluctuations, and ripple remained within acceptable bounds. The large inductor played a key role in stabilizing current flow, while the capacitor dampened output noise.

## Conclusion

This project demonstrated the effective design of a 72V to 12V buck converter tailored for EV applications. With careful component selection, control logic, and simulation analysis, the system proved capable of providing reliable power to vehicle accessories under real-world condition.